

Ngoc Minh Quang Nguyen

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EDUCATION

Carnegie Mellon University PhD, Statistics & Data Science	2022 – present
Loyola University Chicago MS, Applied Statistics	2020 – 2021
Wittenberg University BS, Mathematics & Data Science	2017 – 2020

PUBLICATIONS

JOURNAL ARTICLES

R. Yurko, **Q. Nguyen**, K. Pelechris. NFL Ghosts: A framework for evaluating defender positioning with conditional density estimation. 2026. *Annals of Applied Statistics* **20**(1), 873–892. [doi:10.1214/25-AOAS2132](#); [arXiv:2406.17220](#).

Q. Nguyen, R. Yurko. A multilevel model with heterogeneous variances for snap timing in the National Football League. 2025. *Journal of the Royal Statistical Society: Series A (Statistics in Society)*. [doi:10.1093/jrssa/qnaf185](#); [arXiv:2502.16313](#).

Q. Nguyen, R. Jiang, M. Ellingwood, R. Yurko. Fractional Tackles: Leveraging Player Tracking Data for Within-Play Tackling Evaluation in American Football. *Scientific Reports (Nature Portfolio)* **15**, 2148. 2025. [doi:10.1038/s41598-025-85993-1](#); [arXiv:2403.14769](#).

Q. Nguyen, R. Yurko, G. J. Matthews. Here Comes the STRAIN: Analyzing Defensive Pass Rush in American Football with Player Tracking Data. *The American Statistician* **78**(2), 199–208. 2024. [doi:10.1080/00031305.2023.2242442](#); [arXiv:2305.10262](#).

Editors' Choice Collection.

Q. Nguyen, G. J. Matthews. Filling the Gaps: A Multiple Imputation Approach to Estimating Aging Curves in Baseball. *Journal of Sports Analytics* **10**(1), 77–85. 2024. [doi:10.3233/JSA-240744](#); [arXiv: 2210.02383](#).

N. M. Lemon, L. K. Taylor, M. A. Rech, **Q. Nguyen**, G. J. Matthews, G. Lew, S. Lovett. Utility of D-dimer in predicting pulmonary embolism in patients with COVID-19 presenting to the emergency department. *Journal of the American College of Emergency Physicians Open* **5**(4), e13237. 2024. [doi:10.1002/emp2.13237](#).

B. S. Baumer, G. J. Matthews, **Q. Nguyen**. Big Ideas in Sports Analytics and Statistical Tools for their Investigation. *Wiley Interdisciplinary Reviews: Computational Statistics* **15**(6), e1612. 2023. [doi:10.1002/wics.1612](#); [arXiv:2301.04001](#).

Q. Nguyen. Poisson Modeling and Predicting English Premier League Goal Scoring. *New England Journal of Statistics in Data Science* **1**(1), 95–101. 2023. [doi:10.51387/21-NEJSDS1](#); [arXiv:2105.09881](#).

Q. Nguyen, H. Butler, G. J. Matthews. An Examination of Olympic Sport Climbing Competition Format and Scoring System. *Journal of Data Science* **20**(2), 156–167. 2022. [doi:10.6339/22-JDS1042](#); [arXiv:2111.05310](#).

PREPRINTS & WORKING PAPERS

Q. Nguyen, R. Yurko. Bayesian multilevel step-and-turn models for evaluating player movement in American football. 2026+. [arXiv:2603.17866](#).

Q. Nguyen, R. Yurko. A Bayesian circular mixed-effects model for explaining variability in directional movement in American football. 2025+. [arXiv:2507.06122](#).

Q. Nguyen, P. Campana, N. M. D. Niezink. Co-evolution of cooperation and conflict among organized crime groups: Evidence from the United Kingdom. 2025+.

OTHER

Q. Nguyen, R. Yurko (collaboration between CMSAC and NFL). Introducing the NFL Draft Tier Chart. NFL Football Operations. 2026. <https://operations.nfl.com/updates/football-ops/introducing-the-nfl-draft-tier-chart>

Q. Nguyen, R. Yurko (collaboration between CMSAC and NFL). Building an NFL Draft Value Chart. NFL Football Operations. 2026. <https://operations.nfl.com/updates/football-ops/building-an-nfl-draft-value-chart>

Q. Nguyen, R. Yurko. Down, set, hut! Explaining variability in snap timing on plays with motion. NFL Big Data Bowl 2025. <https://www.kaggle.com/code/tindata/down-set-hut>

Q. Nguyen, L. Jiang, M. Ellingwood, R. Yurko. Momentum-based fractional tackles. NFL Big Data Bowl 2024. <https://www.kaggle.com/code/tindata/momentum-based-fractional-tackles>

G. J. Matthews, **Q. Nguyen**. STRAIN: Sacks, Tackles, Rushing, Aggression INDEX. NFL Big Data Bowl 2023. <https://www.kaggle.com/code/statsinthewild/strain-sacks-tackles-rushing-aggression-index>

B. S. Baumer, **Q. Nguyen**, G. J. Matthews. CRAN Task View: Sports Analytics. The Comprehensive R Archive Network. 2022. <https://cran.r-project.org/web/views/SportsAnalytics.html>

ACADEMIC EXPERIENCE

INSTRUCTOR

Carnegie Mellon University

36-315: Statistics Graphics and Visualization

Summer 2026

Summer Undergraduate Research Experience in Statistics

Summer 2024, Summer 2025

Loyola University Chicago

STAT 103: Fundamentals of Statistics (3 sections)

Spring 2022

GRADUATE TEACHING ASSISTANT

Carnegie Mellon University

36-460/660: Sports Analytics

Spring 2024, Spring 2025, Spring 2026

36-468/668: Text Analysis

Fall 2023, Fall 2024, Fall 2025

Summer Undergraduate Research Experience in Statistics

Summer 2023

36-402/608: Advanced Methods for Data Analysis

Spring 2023

36-401/607: Modern Regression

Fall 2022

OTHER

Wittenberg University

Math Workshop Head Tutor

2018 – 2020

Faculty Aide

2018 – 2020

Clarkston Senior High School

S.U.M. (Student Understanding Math) Assistant

2017

RESEARCH EXPERIENCE

Carnegie Mellon University

2023 – present

Graduate Researcher (Advisor: Ron Yurko)

Carnegie Mellon University

2022 – 2023

Advanced Data Analysis Project (Advisor: Nynke Niezink)

Loyola University Chicago

2020 – 2021

Graduate Research Assistant (Advisor: Gregory J. Matthews)

Wittenberg University

2019 – 2020

Senior Honors Thesis Research (Advisor: Douglas M. Andrews)

PRESENTATIONS

TALKS

Q. Nguyen, R. Yurko. A Bayesian circular mixed-effects model for evaluating change of direction in American football. *Joint Statistical Meetings*. Boston, MA. August 2026. (Contributed)

Q. Nguyen, R. Yurko. NFL CoD: A Bayesian circular mixed-effects model for explaining variability in directional movement in American football. *New England Symposium on Statistics in Sports*. Cambridge, MA. September 2025. (Invited)

Q. Nguyen. Advances in pass rush and change of direction evaluation in American football with player tracking data. *U.S. Olympic and Paralympic Performance Innovation Summit*. Colorado Springs, CO. August 2025. (Invited)

Q. Nguyen. NFL CoD. *Uncontrolled Variables: A Science & Comedy Show*. Chicago, IL. August 2025. (Invited)

Q. Nguyen, R. Yurko. Down, set, hut! Explaining variability in snap timing on plays with motion. *Joint Statistical Meetings*. Nashville, TN. August 2025. (Contributed)

Q. Nguyen, R. Yurko. Step-and-turn modeling of individual player movement in American football. *Conference on Predictive Inference in Sports*. Ames, IA. June 2025. (Invited)

Q. Nguyen, R. Yurko. Explaining variability in player turning behavior in the National Football League. *CMSAC Football Analytics Workshop*. Pittsburgh, PA. May 2025. (Invited)

Q. Nguyen. Advances in pass rush and tackling evaluation in American football with player tracking data. *2024 Joint Statistical Meetings*. Portland, OR. August 2024. (Invited)

Q. Nguyen, L. Jiang, M. Ellingwood, R. Yurko. Fractional tackles. *UConn Sports Analytics Symposium*. Storrs, CT. April 2024. (Invited)

Q. Nguyen, L. Jiang, M. Ellingwood, R. Yurko. Momentum-based fractional tackles. *NFL Big Data Bowl 2024*. Indianapolis, IN. February 2024. (Invited)

Finalist (Top 5 + \$12,500 prize + Presented at 2024 NFL Scouting Combine)

Q. Nguyen. Opportunities in sports analytics with player tracking data: A case study of NFL pass rush evaluation. *Data Science Seminar, Loyola University Chicago*. Chicago, IL. November 2023. (Invited)

Q. Nguyen, R. Yurko, G. J. Matthews. Here Comes the STRAIN: Analyzing Defensive Pass Rush in American Football with Player Tracking Data. *New England Symposium on Statistics in Sports*. Cambridge, MA. September 2023. (Invited, featured talk)

G. J. Matthews, **Q. Nguyen**. STRAIN: Sacks, Tackles, Rushing, Aggression INdex. *NFL Big Data Bowl 2023*. Indianapolis, IN. March 2023. (Invited)

Finalist (Top 8 + \$10,000 prize + Presented at 2023 NFL Scouting Combine)

Q. Nguyen, G. J. Matthews. Estimating Aging Curves: Using Multiple Imputation to Examine Career Trajectories of MLB Offensive Players. *Carnegie Mellon Sports Analytics Conference*. Pittsburgh, PA. October 2022. (Invited)

Finalist, Open Methodology Track, Reproducible Research Competition

Q. Nguyen, H. Butler, G. J. Matthews. An Examination of Olympic Sport Climbing Competition Format and Scoring System. *Cascadia Symposium on Statistics in Sports*. Vancouver, BC. September 2022. (Invited)

Q. Nguyen, H. Butler, G. J. Matthews. An Examination of Sport Climbing's Competition Format and Scoring System. *Carnegie Mellon Sports Analytics Conference*. Pittsburgh, PA. November 2021. (Invited, virtual talk)

Winner, Open Methodology Track, Reproducible Research Competition

POSTERS

Q. Nguyen, H. Butler, G. J. Matthews. Concerns Regarding Sport Climbing Competition Format and Scoring System: Who's Suffering? *UConn Sports Analytics Symposium*. October 2021. (Virtual)

Winner, Student Poster Award

Q. Nguyen. Poisson Modeling and Predicting English Premier League Goal Scoring. *NextGen: Data Science Day, New England Statistical Society*. November 2020. (Virtual)

Co-Winner, Poster Competition, Undergrad/MS Group

Q. Nguyen. Poisson Modeling and Predicting English Premier League Goal Scoring. *Carnegie Mellon Sports Analytics Conference*. October 2020. (Virtual)

Q. Nguyen. Poisson Modeling and Predicting English Premier League Goal Scoring. *UConn Sports Analytics Symposium*. October 2020. (Virtual)

TUTORIALS & WORKSHOPS

Q. Nguyen. Unsupervised learning with tracking data (Advanced level). *Connecticut Sports Analytics Symposium*. Storrs, CT. April 2026.

Q. Nguyen. Rcpp: A primer. *StatBytes Statistical Computing Seminars (Department of Statistics & Data Science, Carnegie Mellon University)*. Pittsburgh, PA. November 2025.

Q. Nguyen. Building a performance metric with player tracking data (Advanced level). *Connecticut Sports Analytics Symposium*. New Haven, CT. April 2025.

Q. Nguyen. Get Trained & Get Strained: The Ingredients of a Big Data Bowl Metric. *Carnegie Mellon Sports Analytics Conference*. Pittsburgh, PA. November 2023.

GUEST LECTURES

Q. Nguyen. Bayesian multilevel modeling in the wild: Beyond homogeneity. *Special Topics: Sports Analytics (36-460/660, Carnegie Mellon University)*. March 2025 & March 2026.

Q. Nguyen. Big Data Bowl: Overview, Getting Started, My Experience, and more. *Summer Undergraduate Research Experience in Statistics (Carnegie Mellon Sports Analytics Camp)*. July 2023.

Q. Nguyen. Web Scraping: A Primer. *Summer Undergraduate Research Experience in Statistics (Carnegie Mellon Sports Analytics Camp)*. July 2023.

HONORS & AWARDS

Sports Analytics Student Award , Carnegie Mellon Statistics & Data Science Funded by the Richardson Matano Family	2025
NFL Big Data Bowl Semifinalist	2025
NFL Big Data Bowl Finalist	2023, 2024
ASA Section on Statistics in Sports Student Funding	2024
Graduate Student Assembly/Provost Conference Funds , Carnegie Mellon University	2024
Student Travel Support , UConn Sports Analytics Symposium	2024
Reproducible Research Competition Finalist , Carnegie Mellon Sports Analytics Conference	2022
University of Toronto Sports Analytics Group Scholarship , Cascadia Symposium on Statistics in Sports	2022
Reproducible Research Competition Winner , Carnegie Mellon Sports Analytics Conference	2021
Student Poster Award Winner , UConn Sports Analytics Symposium	2021
Merit Award , MS in Applied Statistics Program, Loyola University Chicago	2020 – 2021
Poster Competition Co-Winner , NextGen: Data Science Day, New England Statistical Society	2020
Summa Cum Laude , Wittenberg University	2020
Departmental Honors , Department of Mathematics and Computer Science, Wittenberg University	2020

Phi Beta Kappa , Omicron of Ohio Chapter, Wittenberg University	2020
Pi Mu Epsilon , Ohio Upsilon Chapter, Wittenberg University	2020
The Charles and Elsie Little Mathematics Award Department of Mathematics and Computer Science, Wittenberg University Awarded to the most outstanding senior majoring in mathematics and data science	2020
Mortar Board , Wittenberg University	2020
Pick and Pen , Wittenberg University	2019
Mathematics 202 Award , Department of Mathematics and Computer Science, Wittenberg University	2018
Textbook Award , Department of Mathematics and Computer Science, Wittenberg University	2018
Phi Eta Sigma , Wittenberg University	2018
Dean's List , Wittenberg University	All semesters; 2017 – 2020

SERVICES

Organizer

Carnegie Mellon Sports Analytics Center (CMSAC) Research Lab	2023 – present
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Program Committee

UConn/Connecticut Sports Analytics Symposium	2024, 2025, 2026
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Reviewer

Journal of Quantitative Analysis in Sports, Machine Learning, The American Statistician, WIREs Computational Statistics

Judge

CMSAC Reproducible Research Competition	2024, 2025
CSAS Student Poster Award	2024, 2025
CSAS Data Challenge	2024, 2025

Mentoring

Women in Data Science (WiDS) Chicago Datathon	2022
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Statistical Consulting

Carnegie Mellon University Athletics	2024
Loyola University Chicago	2021 – 2022
(Clients: Stritch School of Medicine; Department of Biology; City of Milwaukee Assessor's Office)	

SKILLS

Computing: R/RStudio (expert); Python, SQL, SAS, Git/GitHub (intermediate)

Typesetting: markdown, L^AT_EX (proficient)

Bilingual: Vietnamese (native language) and English (proficient)

Last updated May 24, 2026